

with homogeneous Neumann boundary conditions on $\partial\Omega$.

Before entering into the details making the above approach rigorous (next section), we want to present some examples of this kind of equations. We will consider two functionals that we already analyzed in Chapter 7, and more precisely

$$\mathcal{F}(\rho) = \int f(\rho(x)) dx, \quad \text{and} \quad \mathcal{V}(\rho) = \int V(x) d\rho.$$

In this case we already saw that we have

$$\frac{\delta \mathcal{F}}{\delta \rho}(\rho) = f'(\rho), \quad \frac{\delta \mathcal{V}}{\delta \rho}(\rho) = V.$$

An interesting example is the case $f(t) = t \log t$. In such a case we have $f'(t) = \log t + 1$ and $\nabla(f'(\rho)) = \frac{\nabla \rho}{\rho}$: this means that the gradient flow equation associated to the functional $\mathcal{F}$ would be the *Heat Equation*

$$\partial_t \rho - \Delta \rho = 0,$$

and that for $\mathcal{F} + \mathcal{V}$ we would have the *Fokker-Planck Equation*

$$\partial_t \rho - \Delta \rho - \nabla \cdot (\rho \nabla V) = 0.$$

We will see a list of other Gradient Flow equations in the Discussion section, including the so-called *Porous Media Equation*, obtained for other choices of f .

Remark 8.4. One could wonder how the usual stochastic diffusion interpretation of the Heat and Fokker-Planck Equations is compatible with the completely deterministic framework that we present here, where the velocity of each particle is given by $-\nabla \rho / \rho - \nabla V$ (take $V = 0$ to get the Heat Equation). Indeed, the Fokker-Planck Equation is also the PDE solved by the density of a bunch of particle following a stochastic equation $dX_t = -\nabla V(X_t)dt + dB_t$, where B is a standard Brownian motion (independent for each particle). The answer, as it is well pointed out in [89], is based on the idea that in this optimal transport interpretation we rearrange (i.e., relabel) the particles. This fact is quite clear in the 1D case: if at every time step particles move from x to $x - \tau \nabla V(x) + B_\tau$ (i.e. they follow the drift $-\nabla V$ and they diffuse with a Gaussian law), and then we reorder them (we always call particle 1 the one which is the most at the left, particle 2 the next one... and particle N the one which is the most at the right), then we have a discrete family of trajectories which converge, when τ goes to 0 and the number N to infinity, to a solution of Fokker-Planck where the velocity is exactly given by $-\nabla \rho / \rho - \nabla V$. Similar considerations can be done in higher dimension with suitable notions of rearrangement.

Note that all these PDEs come accompanied by Neumann boundary conditions $\rho \frac{\partial}{\partial \mathbf{n}} \left(\frac{\delta F}{\delta \rho}(\rho) \right) = 0$ on $\partial\Omega$, as a consequence of the Neumann boundary conditions for the continuity equation of Section 5.3. We will see in Section 8.4.3 a case of extension to Dirichlet boundary conditions.

We finish this section with some philosophical thoughts. Why study some PDEs considering them as gradient flows for the distance W_2 ? There are at least three reasons.

The first one is that it allows to give an existence result (of a weak solution of such a PDE), with the technique that we will see in the next section. This is obviously useless when the equation is already well-known, as it is the case for the Heat equation, or its Fokker-Planck variant. But the same strategy could be applied to variants of the same equations (or to similar equations in stranger spaces, as it is nowadays done for the Heat flow in metric measure spaces, see [184, 18]). It is also very useful when the equation is of new type (we will see the case of the crowd motion models in Section 8.4.2). The second goal could be to derive properties of the flow and of the solution, once we know that it is a gradient flow. A simple one is the fact that $t \mapsto F(\rho_t)$ must be decreasing in time. However, this can be often deduced from the PDE in different ways and does not require in general to have a gradient flow. The third goal concerns numerics: the discretized scheme to get a gradient flow is itself a sort of algorithm to approximate the solution. If the minimization problems at each time step are discretized and attacked in a suitable way, this can provide efficient numerical tools (see, for instance, [41]).

Finally, all the procedure we presented is related to the study and the existence of weak solutions. What about uniqueness? We stress that in PDE applications the important point is to prove uniqueness for weak solutions of the continuity equation with $\mathbf{v} = -\nabla \frac{\delta F}{\delta \rho}$ (i.e. we do not care at the metric structure, and at the definitions of EVI and EDE). In some cases, this uniqueness could be studied independently of the gradient-flow structure (this is the case for the Heat equation, for instance). Anyway, should we use PDE approaches for weak solutions, or abstract approaches in metric spaces, it turns out that usually uniqueness is linked to some kind of convexity, or λ -convexity, and in particular to displacement convexity. This is why a large part of the theory has been developed for λ -geodesically convex functionals.

8.3 Analysis of the Fokker-Planck case

We consider here the case study of the Fokker-Planck equation, which is the gradient flow of the functional

$$J(\rho) = \int_{\Omega} \rho \log \rho + \int_{\Omega} V d\rho,$$

where V is a Lipschitz function on the compact domain Ω . The initial measure $\rho_0 \in \mathcal{P}(\Omega)$ is taken such that $J(\rho_0) < +\infty$.

We stress that the first term of the functional is defined as

$$\mathcal{F}(\rho) := \begin{cases} \int_{\Omega} \rho(x) \log \rho(x) dx & \text{if } \rho \ll \mathcal{L}^d, \\ +\infty & \text{otherwise,} \end{cases}$$

where we identify the measure ρ with its density, when it is absolutely continuous. This functional is l.s.c. thanks to Proposition 7.7 (which can be applied since we are on a domain with finite measure, otherwise the situation is trickier, and we refer to Ex(45)).

semi-continuity allows to establish the following

Proposition 8.5. *The functional J has a unique minimum over $\mathcal{P}(\Omega)$. In particular J is bounded from below. Moreover, for each $\tau > 0$ the following sequence of optimization*

problems recursively defined is well-posed

$$\rho_{(k+1)}^\tau \in \operatorname{argmin}_\rho J(\rho) + \frac{W_2^2(\rho, \rho_{(k)}^\tau)}{2\tau}, \quad (8.5)$$

which means that there is a minimizer at every step, and this minimizer is unique.

Proof. Just apply the direct method, noting that $\mathcal{P}(\Omega)$ is compact for the weak convergence, which is the same as the convergence for the W_2 distance (again, because Ω is compact), and for this convergence $\mathcal{F}$ is l.s.c. and the other terms are continuous. This gives at the same time the existence of a minimizer for J and of a solution to each of the above minimization problems (8.5). Uniqueness comes from the fact that all the functionals are convex (in the usual sense) and $\mathcal{F}$ is strictly convex. $\square$ $\square$

Optimality conditions at each time step We will use Propositions 7.17, 7.20 and 7.18, and to do so we first need to prove the following result.

Lemma 8.6. *Any minimizer $\hat{\rho}$ in (8.5) must satisfy $\hat{\rho} > 0$ a.e. and $\log \hat{\rho} \in L^1$.*

Proof. Consider the measure $\tilde{\rho}$ with constant positive density c in Ω (i.e. $c = |\Omega|^{-1}$). Let us define ρ_ε as $(1 - \varepsilon)\hat{\rho} + \varepsilon\tilde{\rho}$ and compare $\hat{\rho}$ to ρ_ε .

By optimality of $\hat{\rho}$, we may write

$$\mathcal{F}(\hat{\rho}) - \mathcal{F}(\rho_\varepsilon) \leq \int_\Omega V d\rho_\varepsilon - \int_\Omega V d\hat{\rho} + \frac{W_2^2(\rho_\varepsilon, \rho_{(k)}^\tau)}{2\tau} - \frac{W_2^2(\hat{\rho}, \rho_{(k)}^\tau)}{2\tau}. \quad (8.6)$$

The Wasserstein term in the right hand side may be estimated by convexity:

$$\frac{W_2^2(\rho_\varepsilon, \rho_{(k)}^\tau)}{2\tau} \leq (1 - \varepsilon) \frac{W_2^2(\hat{\rho}, \rho_{(k)}^\tau)}{2\tau} + \varepsilon \frac{W_2^2(\tilde{\rho}, \rho_{(k)}^\tau)}{2\tau}.$$

The potential term is of order ε as well:

$$\int_\Omega V d\rho_\varepsilon - \int_\Omega V d\hat{\rho} = \varepsilon \int_\Omega V d(\tilde{\rho} - \hat{\rho}) \leq C\varepsilon.$$

This shows that the whole right hand side of (8.6) is estimated by $C\varepsilon$ and we get

$$\int_\Omega f(\hat{\rho}) - f(\rho_\varepsilon) \leq C\varepsilon$$

where $f(t) = t \log t$ (set to 0 in $t = 0$). Write

$$A = \{x \in \Omega : \hat{\rho}(x) > 0\}, \quad B = \{x \in \Omega : \hat{\rho}(x) = 0\}.$$

Since f is convex we write, for $x \in A$, $f(\hat{\rho}(x)) - f(\rho_\varepsilon(x)) \geq (\hat{\rho}(x) - \rho_\varepsilon(x))f'(\rho_\varepsilon(x)) = \varepsilon(\hat{\rho}(x) - \tilde{\rho}(x))(1 + \log \rho_\varepsilon(x))$. For $x \in B$ we simply write $f(\hat{\rho}(x)) - f(\rho_\varepsilon(x)) = -\varepsilon c \log(\varepsilon c)$. This allows to write

$$-\varepsilon c \log(\varepsilon c)|B| + \varepsilon \int_A (\hat{\rho}(x) - c)(1 + \log \rho_\varepsilon(x)) dx \leq C\varepsilon$$

and, dividing by ε ,

$$-c \log(\varepsilon c)|B| + \int_A (\widehat{\rho}(x) - c)(1 + \log \rho_\varepsilon(x)) dx \leq C \quad (8.7)$$

Note that the always have

$$(\widehat{\rho}(x) - c)(1 + \log \rho_\varepsilon(x)) \geq (\widehat{\rho}(x) - c)(1 + \log c)$$

(just distinguish between the case $\widehat{\rho}(x) \geq c$ and $\widehat{\rho}(x) \leq c$). Thus, we may write

$$-c \log(\varepsilon c)|B| + \int_A (\widehat{\rho}(x) - c)(1 + \log c) dx \leq C.$$

Letting $\varepsilon \rightarrow 0$ provides a contradiction, unless $|B| = 0$.

This proves $\widehat{\rho} > 0$ a.e. We now come back to (8.7), which is an upper bound on the integral of the functions $(\widehat{\rho}(x) - c)(1 + \log \rho_\varepsilon(x))$. We already noted that these functions are bounded from below by $(\widehat{\rho}(x) - c)(1 + \log c)$, which is L^1 because c is a constant and $\widehat{\rho} \in L^1$. Hence, we can apply Fatou's lemma and obtain, at the limit as $\varepsilon \rightarrow 0$,

$$\int_\Omega (\widehat{\rho}(x) - c)(1 + \log \widehat{\rho}(x)) dx \leq C$$

(the integral on A has been replaced by the integral on Ω , since B is negligible). Since we already know that $(\widehat{\rho} - c)(1 + \log \widehat{\rho})$ is bounded from below by an L^1 function, and its integral is finite, then it is L^1 . But we already know that $\widehat{\rho}, \widehat{\rho} \log \widehat{\rho} \in L^1$, we deduce $\log \widehat{\rho} \in L^1$. $\square$ $\square$

We can now compute the first variation and give optimality conditions on the optimal $\rho_{(k+1)}^\tau$.

Proposition 8.7. *The optimal measure $\rho_{(k+1)}^\tau$ in (8.5) satisfies*

$$\log(\rho_{(k+1)}^\tau) + V + \frac{\bar{\varphi}}{\tau} = \text{constant a.e.} \quad (8.8)$$

where $\bar{\varphi}$ is the (unique) Kantorovich potential from $\rho_{(k+1)}^\tau$ to $\rho_{(k)}^\tau$. In particular, $\log \rho_{(k+1)}^\tau$ is Lipschitz continuous. If T_k^τ is the optimal transport from $\rho_{(k+1)}^\tau$ to $\rho_{(k)}^\tau$, then it satisfies

$$\mathbf{v}_{(k)}^\tau := \frac{\text{id} - T_k^\tau}{\tau} = -\nabla \left(\log(\rho_{(k+1)}^\tau) + V \right) \text{ a.e.} \quad (8.9)$$

Proof. Take the optimal measure $\widehat{\rho} := \rho_{(k+1)}^\tau$. We can say that it is regular (see Definition 7.12) for the functional J , since the only term which is not finite on all probabilities in $\mathcal{P}(\Omega)$ is the entropy term, but it is convex and finite both on the optimum $\widehat{\rho}$ and on L^∞ densities. We can check that J admits a first variation at $\widehat{\rho}$. For the linear term it is straightforward, and for the Wasserstein term we can apply Proposition 7.17. The uniqueness of the Kantorovich potential is guaranteed by Proposition 7.18 together with Lemma 8.6. For the entropy term, consider $\rho_\varepsilon := (1 - \varepsilon)\widehat{\rho} + \varepsilon\tilde{\rho}$, for $\tilde{\rho} \in L^\infty$.

Set $M = \|\tilde{\rho}\|_{L^\infty}$ and look at $\varepsilon \mapsto \int \rho_\varepsilon \log \rho_\varepsilon$. The integrand can be differentiated in ε pointwisely, thus getting $(1 + \log \rho_\varepsilon)(\tilde{\rho} - \hat{\rho})$. For $\varepsilon < 1/2$, we can check that these functions are dominated by $(\hat{\rho} + M)(|\log \hat{\rho}| + \log M + 1)$, which is L^1 thanks to $\hat{\rho} \in L^1$ and $\log \hat{\rho} \in L^1$ (from Lemma 8.6). This allows to differentiate under the integral sign and proves that the first variation of this term is $1 + \log \hat{\rho}$.

The first variation of J is hence

$$\frac{\delta J}{\delta \rho} = f'(\rho) + V = \log(\rho) + 1 + V.$$

Applying Proposition 7.20 we obtain Equation (8.8), which is valid a.e. since $\hat{\rho} > 0$. In particular, this implies that $\rho_{(k+1)}^\tau$ is Lipschitz continuous, since we have

$$\rho_{(k+1)}^\tau(x) = \exp\left(C - V(x) - \frac{\bar{\varphi}(x)}{\tau}\right).$$

Then, one differentiates and gets the equality

$$\nabla \bar{\varphi} = \frac{\text{id} - T_k^\tau}{\tau} = -\nabla\left(\log(\rho_{(k+1)}^\tau) + V\right) \text{ a.e.}$$

and this allows to conclude. $\square$

Interpolation between time steps and uniform estimates Let us collect some other tools

Proposition 8.8. *For any $\tau > 0$, the sequence of minimizers satisfies*

$$\sum_k \frac{W_2^2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{\tau} \leq C := 2(J(\rho_0) - \inf J).$$

Proof. This is obtained by comparing the optimizer $\rho_{(k+1)}^\tau$ to the previous measure $\rho_{(k)}^\tau$. We get

$$J(\rho_{(k+1)}^\tau) + \frac{W_2^2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{2\tau} \leq J(\rho_k^\tau), \quad (8.10)$$

which implies

$$\sum_k \frac{W_2^2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{\tau} \leq \sum_k 2(J(\rho_k^\tau) - J(\rho_{(k+1)}^\tau)).$$

This last sum is telescopic and gives the claim. $\square$

Let us define two interpolations between the measures $\rho_{(k)}^\tau$.

With this time-discretized method, we have obtained, for each $\tau > 0$, a sequence $(\rho_{(k)}^\tau)_k$. We can use it to build at least two interesting curves in the space of measures:

- first we can define some piecewise constant curves, i.e. $\rho_t^\tau := \rho_{(k+1)}^\tau$ for $t \in]k\tau, (k+1)\tau]$; associated to this curve we also define the velocities $\mathbf{v}_t^\tau = \mathbf{v}_{(k+1)}^\tau$ for $t \in]k\tau, (k+1)\tau]$, where $\mathbf{v}_{(k)}^\tau$ is defined as in (8.9): $\mathbf{v}_{(k)}^\tau = (\text{id} - T_k^\tau)/\tau$, taking as T_k^τ the optimal transport from $\rho_{(k+1)}^\tau$ to $\rho_{(k)}^\tau$; we also define the momentum variable $E^\tau = \rho^\tau \mathbf{v}^\tau$;
- then, we can also consider the densities $\tilde{\rho}_t^\tau$ that interpolate the discrete values $(\rho_{(k)}^\tau)_k$ along geodesics:

$$\tilde{\rho}_t^\tau = \left(\frac{k\tau - t}{\tau} \mathbf{v}_{(k)}^\tau + \text{id} \right)_\# \rho_{(k)}^\tau, \text{ for } t \in](k-1)\tau, k\tau[; \quad (8.11)$$

the velocities $\tilde{\mathbf{v}}_t^\tau$ are defined so that $(\tilde{\rho}^\tau, \tilde{\mathbf{v}}^\tau)$ satisfy the continuity equation and $\|\tilde{\mathbf{v}}_t^\tau\|_{L^2(\tilde{\rho}_t^\tau)} = |(\tilde{\rho}^\tau)'(t)|$. To do so, we take

$$\tilde{\mathbf{v}}_t^\tau = \mathbf{v}_t^\tau \circ \left((k\tau - t) \mathbf{v}_{(k)}^\tau + \text{id} \right)^{-1};$$

as before, we define a momentum variable: $\tilde{E}_\tau = \tilde{\rho}^\tau \tilde{\mathbf{v}}^\tau$.

After these definitions we consider some a priori bounds on the curves and the velocities that we defined. We start from some estimates which are standard in the framework of Minimizing Movements.

Note that the velocity (i.e. metric derivative) of $\tilde{\rho}^\tau$ is constant on each interval $]k\tau, (k+1)\tau[$ and equal to

$$\frac{W_2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{\tau} = \frac{1}{\tau} \left(\int |\text{id} - T_k^\tau|^2 d\rho_{(k+1)}^\tau \right)^{1/2} = \|\mathbf{v}_{k+1}^\tau\|_{L^2(\rho_{(k+1)}^\tau)},$$

which gives

$$\|\tilde{\mathbf{v}}_t^\tau\|_{L^2(\tilde{\rho}_t^\tau)} = |(\tilde{\rho}^\tau)'(t)| = \frac{W_2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{\tau} = \|\mathbf{v}_t^\tau\|_{L^2(\rho_t^\tau)},$$

where we used the fact that the velocity field $\tilde{\mathbf{v}}^\tau$ has been chosen so that its L^2 norm equals the metric derivative of the curve $\tilde{\rho}^\tau$.

In particular we can obtain

$$\begin{aligned} |E^\tau|([0, T] \times \Omega) &= \int_0^T dt \int_\Omega |\mathbf{v}_t^\tau| d\rho_t^\tau = \int_0^T \|\mathbf{v}_t^\tau\|_{L^1(\rho_t^\tau)} dt \leq \int_0^T \|\mathbf{v}_t^\tau\|_{L^2(\rho_t^\tau)} dt \\ &\leq T^{1/2} \int_0^T \|\mathbf{v}_t^\tau\|_{L^2(\rho_t^\tau)}^2 dt = T^{1/2} \sum_k \tau \left(\frac{W_2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{\tau} \right)^2 \leq C. \end{aligned}$$

The estimate on $\tilde{E}^\tau$ is completely analogous

$$\begin{aligned} |\tilde{E}^\tau|([0, T] \times \Omega) &= \int_0^T dt \int_\Omega |\tilde{\mathbf{v}}_t^\tau| d\tilde{\rho}_t^\tau \leq T^{1/2} \int_0^T \|\tilde{\mathbf{v}}_t^\tau\|_{L^2(\tilde{\rho}_t^\tau)}^2 dt \\ &= T^{1/2} \sum_k \tau \left(\frac{W_2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{\tau} \right)^2 \leq C. \end{aligned}$$

This gives compactness of E^τ and $\tilde{E}^\tau$ in the space of vector measures on space-time, for the weak convergence. As far as $\tilde{\rho}^\tau$ is concerned, we can obtain more than that. Consider the following estimate, for $s < t$

$$W_2(\tilde{\rho}_t^\tau, \tilde{\rho}_s^\tau) \leq \int_s^t |(\tilde{\rho}^\tau)'|(r) dr \leq (t-s)^{1/2} \left(\int_s^t |(\tilde{\rho}^\tau)'|(r)^2 dr \right)^{1/2}.$$

From the previous computations, we have again

$$\int_0^T |(\tilde{\rho}^\tau)'|(r)^2 dr = \sum_k \tau \left(\frac{W_2(\rho_{(k+1)}^\tau, \rho_{(k)}^\tau)}{\tau} \right)^2 \leq C,$$

and this implies

$$W_2(\tilde{\rho}_t^\tau, \tilde{\rho}_s^\tau) \leq C(t-s)^{1/2}, \quad (8.12)$$

which means that the curves $\tilde{\rho}^\tau$ are uniformly Hölder continuous. Since they are defined on $[0, T]$ and valued in $\mathbb{W}_2(\Omega)$ which is compact, we can apply the Ascoli Arzelà Theorem. This implies that, up to subsequences, we have

$$\begin{aligned} E^\tau &\rightharpoonup E \text{ in } \mathcal{M}^d([0, T] \times \Omega), \quad \tilde{E}^\tau \rightharpoonup \tilde{E} \text{ in } \mathcal{M}^d([0, T] \times \Omega); \\ \tilde{\rho}^\tau &\rightarrow \rho \text{ uniformly for the } W_2 \text{ distance.} \end{aligned}$$

The limit curve ρ , from the uniform bounds on $\tilde{\rho}^\tau$, is both $\frac{1}{2}$ -Hölder continuous and absolutely continuous in $\mathbb{W}_2$.

As far as the curves ρ^τ are concerned, they also converge uniformly to the same curve ρ , since $W_2(\rho_t^\tau, \tilde{\rho}_t^\tau) \leq C\sqrt{\tau}$ (a consequence of (8.12), of the fact that $\tilde{\rho}^\tau = \rho^\tau$ on the points of the form $k\tau$ and of the fact that ρ^τ is constant on each interval $]k\tau, (k+1)\tau[$).

Let us now prove that $\tilde{E} = E$.

Lemma 8.9. *Suppose that we have two families of vector measures E^τ and $\tilde{E}^\tau$ such that*

- $\tilde{E}^\tau = \tilde{\rho}^\tau \tilde{\mathbf{v}}^\tau; E^\tau = \rho^\tau \mathbf{v}^\tau;$
- $\tilde{\mathbf{v}}_t^\tau = \mathbf{v}_t^\tau \circ \left((k\tau - t)\mathbf{v}_{(k)}^\tau + \text{id} \right)^{-1}; \tilde{\rho}^\tau = \left((k\tau - t)\mathbf{v}_{(k)}^\tau + \text{id} \right)_\# \rho^\tau;$
- $\int |v^\tau|^2 d\rho^\tau \leq C$ (with C independent of τ);
- $E^\tau \rightharpoonup E$ and $\tilde{E}^\tau \rightharpoonup \tilde{E}$ as $\tau \rightarrow 0$

Then $\tilde{E} = E$.

Proof. It is sufficient to fix a Lipschitz function $f : [0, T] \times \Omega \rightarrow \mathbb{R}^d$ and to prove $\int f \cdot dE = \int f \cdot d\tilde{E}$. To do that, we write

$$\int f \cdot d\tilde{E}^\tau = \int_0^T dt \int_\Omega f \cdot \tilde{\mathbf{v}}_t^\tau d\tilde{\rho}^\tau = \int_0^T dt \int_\Omega f \circ ((k\tau - t)\mathbf{v}^\tau + \text{id}) \cdot \mathbf{v}_t^\tau d\rho^\tau,$$

which implies

$$\begin{aligned} \left| \int f \cdot d\tilde{E}^\tau - \int f \cdot dE^\tau \right| &\leq \int_0^T dt \int_\Omega |f \circ ((k\tau - t)\mathbf{v}^\tau + \text{id}) - f| |\mathbf{v}_t^\tau| d\rho^\tau \\ &\leq \text{Lip}(f) \tau \int_0^T \int_\Omega |\mathbf{v}_t^\tau|^2 d\rho^\tau \leq C\tau. \end{aligned}$$

This estimate proves that the limit of $\int f \cdot d\tilde{E}^\tau$ and $\int f \cdot dE^\tau$ is the same, i.e. $E = \tilde{E}$. $\square$ $\square$

Relation between ρ and E We can obtain the following:

Proposition 8.10. *The pair (ρ, E) satisfies, in distributional sense*

$$\partial_t \rho + \nabla \cdot E = 0, \quad E = -\nabla \rho - \rho \nabla V,$$

with homogeneous Neumann boundary conditions on $\partial\Omega$. In particular we have found a solution to

$$\begin{cases} \partial_t \rho - \Delta \rho + \nabla \cdot (\rho \nabla V) = 0, \\ \rho(0) = \rho_0, \end{cases}$$

where the initial datum is to be intended in the following sense: the curve $t \mapsto \rho_t$ is (absolutely) continuous in $\mathbb{W}_2$, and its initial value is ρ_0 .

Proof. First, consider the weak convergence $(\tilde{\rho}^\tau, \tilde{E}^\tau) \rightharpoonup (\rho, E)$ (which is a consequence of $\tilde{E} = E$). The continuity equation $\partial_t \tilde{\rho}^\tau + \nabla \cdot \tilde{E}^\tau = 0$ is satisfied (as in Section 4.1.2) in the sense of distributions by $(\tilde{\rho}^\tau, \tilde{E}^\tau)$ (we test the equations against C_c^1 function on $]0, T[\times \Omega$, which means in particular that we do not require the test functions to vanish on $\partial\Omega$) and this passes to the limit.

Hence, $\partial_t \rho + \nabla \cdot E = 0$.

The continuity in $\mathbb{W}_2$ and the initial datum pass to the limit because of the uniform $C^{0,1/2}$ bound in (8.12).

Then, use the convergence $(\rho^\tau, E^\tau) \rightharpoonup (\rho, E)$. Actually, using the optimality conditions of Proposition 8.7 and the definition of $E^\tau = \mathbf{v}^\tau \rho^\tau$, we have, for each $\tau > 0$, $E^\tau = -\nabla \rho^\tau - \rho^\tau \nabla V$ (again, in the sense of distributions: the term $\rho^\tau \nabla V$ is the product of an L^1 and an L^∞ functions). It is not difficult to pass this condition to the limit either. Take $f \in C_c^1(]0, T[\times \Omega; \mathbb{R}^d)$ and test:

$$\int f \cdot dE^\tau = - \int f \cdot \nabla \rho^\tau - \int f \cdot \nabla V \rho^\tau = \int \nabla \cdot f d\rho^\tau - \int f \cdot \nabla V \rho^\tau.$$

These terms pass to the limit as $\rho^\tau \rightharpoonup \rho$, at least if $V \in C^1$, since all the test functions above are continuous. This would give $\int f \cdot dE = \int (\nabla \cdot f) d\rho - \int f \cdot \nabla V d\rho$, which implies $E = -\nabla \rho - \rho \nabla V$ (once more, $\nabla \rho$ is to be intended in the sense of distributions).

To handle the case where V is only Lipschitz continuous, let us note that for every τ, t we have $J(\rho_t^\tau) \leq J(\rho_0)$ (this is a consequence of (8.10), which iterated over k , gives

$J(\rho_k^\tau) \leq J(\rho_0)$). This gives a uniform bound on $\mathcal{F}(\rho_t^\tau)$ and the Dunford-Pettis Theorem below turns the weak convergence $\rho_t^\tau \rightharpoonup \rho_t$ as measures into a weak convergence in L^1 . Look at the term

$$\int f \cdot \nabla V \rho^\tau := \int_0^T dt \int_{\Omega} f(t, x) \cdot \nabla V(x) \rho_t^\tau(x) dx.$$

The integrand in t is uniformly bounded by $\|f\|_{L^\infty} \|\nabla V\|_{L^\infty}$. Hence, we can compute its limit as $\tau \rightarrow 0$ looking at the limit for each fixed t because it is dominated. And, for fixed t , we have weak convergence in L^1 of ρ_t^τ to ρ_t , which, multiplied times a fixed L^∞ function, i.e. $f \cdot \nabla V$, gives the desired limit of the integral. $\square$ $\square$

Box 8.2. – Memo – Equi-integrability and Dunford-Pettis Theorem

Definition - A sequence of function ρ_n is said to be equi-integrable if for every $\varepsilon > 0$ there exists $\delta > 0$ such that on every set A with $|A| \leq \delta$ we have $\int_A \rho_n(x) dx \leq \varepsilon$ for any n .

We can check that equi-integrability is equivalent to a bound $\int f(\rho_n(x)) dx \leq C < +\infty$ for any superlinear function f . Let us only prove that such a bound implies equi-integrability. Indeed, fix $\varepsilon > 0$ and take M such that for all t with $t > M$ we have $f(t)/t > 2C/\varepsilon$ (this is possible thanks to the superlinearity of f). Then we estimate

$$\int_A \rho_n = \int_{A \cap \{\rho_n \leq M\}} \rho_n + \int_{A \cap \{\rho_n > M\}} \frac{\rho_n}{f(\rho_n)} f(\rho_n) dx \leq M|A| + \frac{\varepsilon}{2C} C.$$

It is now enough to take $\delta = \frac{\varepsilon}{2M}$ and we get $\int_A \rho_n(x) dx < \varepsilon$.

Theorem - Suppose that ρ_n is a sequence of probability densities weakly converging as measures to $\rho \in \mathcal{P}(\Omega)$. Suppose that $\int f(\rho_n)$ is bounded for some superlinear function f . Then ρ is also absolutely continuous and the weak convergence also holds in L^1 (i.e. in duality with L^∞ functions, and not only with continuous ones).

Proof - The absolute continuity of ρ is a consequence of the lower semi-continuity of $\mathcal{F}$ defined as $\mathcal{F}(\rho) = \int_{\Omega} f(\rho(x)) dx$ if $\rho \ll \mathcal{L}^d$ ($+\infty$ otherwise).

Moreover, the bound on $\int f(\rho_n(x)) dx$ implies equi-integrability. Fix a test function $\varphi \in L^\infty$ and use the fact that for every $\varepsilon > 0$ there is a $\delta > 0$ with $|A| \leq \delta \Rightarrow \int_A \rho_n(x) dx \leq \varepsilon$ and that, by Lusin Theorem, for this $\delta > 0$ there exists a continuous function $\tilde{\varphi}$ with $\|\tilde{\varphi}\|_{L^\infty} \leq \|\varphi\|_{L^\infty}$ and $\{\varphi \neq \tilde{\varphi}\} \subset A$, where A is open with $|A| < \delta$. We have

$$\left| \int \varphi(\rho_n - \rho) \right| \leq \left| \int \tilde{\varphi}(\rho_n - \rho) \right| + 2\|\varphi\|_{L^\infty} \int_A (\rho_n + \rho).$$

Since A is open and $\rho_n \rightharpoonup \rho$ we have $\int_A \rho(x) dx \leq \liminf \int_A \rho_n(x) dx \leq \varepsilon$, hence

$$\limsup \left| \int \varphi(x) \rho_n(x) dx - \int \varphi(x) \rho(x) dx \right| \leq 0 + 2\|\varphi\|_{L^\infty} \varepsilon,$$

which implies, ε being arbitrary, $\int \varphi(x) \rho_n(x) dx \rightarrow \int \varphi(x) \rho(x) dx$.

8.4 Discussion

8.4.1 EVI, uniqueness and geodesic convexity

In the general theory of Gradient Flows ([17]) in metric spaces, another characterization, different from the EDE, is proposed in order to cope with uniqueness and stability results. It is based on the following observation: if $F : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, then the inequality

$$F(y) \geq F(x) + p \cdot (y - x) \quad \text{for all } y \in \mathbb{R}^d$$

characterizes (by definition) the vectors $p \in \partial F(x)$ and, if $F \in C^1$, it is only satisfied for $p = \nabla F(x)$. Analogously, if F is λ -convex, the inequality that characterizes the gradient is

$$F(y) \geq F(x) + \frac{\lambda}{2} |x - y|^2 + p \cdot (y - x) \quad \text{for all } y \in \mathbb{R}^d.$$

Hence, we can pick a curve $x(t)$ and a point y and compute

$$\frac{d}{dt} \frac{1}{2} |x(t) - y|^2 = (y - x(t)) \cdot (-x'(t)).$$

Consequently, imposing

$$\frac{d}{dt} \frac{1}{2} |x(t) - y|^2 \leq F(y) - F(x(t)) - \frac{\lambda}{2} |x(t) - y|^2,$$

for all y , will be equivalent to $-x'(t) = \nabla F(x(t))$. This will provide a second characterization (called EVI, *Evolution Variational Inequality*) of gradient flows in a metric environment. Indeed, all the terms appearing in the above inequality have a metric counterpart (only squared distances and derivatives w.r.t. time appear). Even if we often forget the dependence on λ , it should be noted that the condition EVI should actually be written as EVI_λ , since it involves a parameter λ , which is a priori arbitrary. Actually, λ -convexity of F is not necessary to define the EVI_λ property, but it will be necessary in order to guarantee the existence of curves which satisfy such a condition. The notion of λ -convexity will hence be crucial also in metric spaces, where it will be rather “ λ -geodesic-convexity”.

The role of the EVI condition in the uniqueness and stability of gradient flows is quite easy to guess. Take two curves, that we call $x(t)$ and $y(s)$, and compute

$$\frac{d}{dt} \frac{1}{2} d(x(t), y(s))^2 \leq F(y(s)) - F(x(t)) - \frac{\lambda}{2} d(x(t), y(s))^2, \quad (8.13)$$

$$\frac{d}{ds} \frac{1}{2} d(x(t), y(s))^2 \leq F(x(t)) - F(y(s)) - \frac{\lambda}{2} d(x(t), y(s))^2. \quad (8.14)$$

If one wants to estimate $E(t) = \frac{1}{2} d(x(t), y(t))^2$, summing up the two above inequalities, after a chain-rule argument for the composition of the function of two variables³ $(t, s) \mapsto \frac{1}{2} d(x(t), y(s))^2$ and of the curve $t \mapsto (t, t)$, gives

$$\frac{d}{dt} E(t) \leq -2\lambda E(t).$$

³Note that a similar argument, based on doubling the variables, is also often performed for the differentiation of W_2^2 along curves in $\mathbb{W}_2$, which we did differently in Section 5.3.5.